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CSA-1501

Class Test

1] Unified Big Data Architecture.

Use a centralized data lake/data warehouse online and offline customer data. Technologies like kafka, spark and hadoop can process data in real-time.

-> Customer analytics and ML models can then provide personalized recommendation,

2] Distributed computing for scientific Data.

Distributed computing divides large data-sets and calculations among multiple machines. Parallel processing reduces execution time, improves resource utilization, and allows the system to scale by adding more nodes compared with a centralized system.

3.] Digital Payment Fraud Detection :-

-> Two use kafka for real-time transaction streams and spark streaming for processing.

-> real time alerts allow suspicious transactions to be blocked quickly.

4.] multimedia Big Data management.

-> video, audio and images require large storage and high processing power.

-> Use cloud/object storage, HDFS, Hadoop and spark for distributed storage and processing.

-> Compression and suitable file formats reduce storage and network costs.

5] Disaster Recovery strategy

-> use multiple servers / availability zones with data replication, backups, redundancy, and automated failover.

-> Technologies such as cloud storage replication and load balancing keep service available even when a system or server fails.

6] Data quality and Governance.

Duplicate, missing and inconsistent data can produce incorrect analytical results. use data validation, deduplication, standardization, metadata management, data ownership and regular data quality checks to improve reliability.

7] Scalable Recommendation system.

-> collect user activities such as searches, views, likes and watch history.

-> use kafka and spark to process large scale behavior data and machine learning algorithms to generate recommendation

-> This improves Personalization and user engagement.

8] Cloud Computing in Big Data.

Cloud Platforms Provide scalability, flexi-bility and Pay-as-you-use Pricing. recources can be increased or decreased as required. However, challenges include security, Privacy. vendor lock-in, network dependency and cost management.

9] Transportation and Logistics Analytics.

→ collect GPS, traffic, vehicle, fuel and delivery data using IoT devices.

→ Kafka and spark can Process data in real time to optimize routes, reduce fuel consumption, Predict delays, and support faster operational decisions.

10] Ethical and legal challenges.

Big data can create risks involving Privacy unauthorized access, bias, and misuse of Personal information. organizations should implement consent Policies, data encryption, access control, anonymiza-tion, audits, transparency and compliance with applicable data Protection laws.